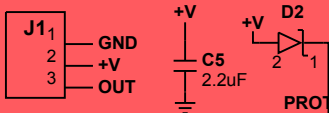


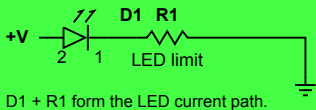
FULL-COLOR FULL SCHEMATIC RETAIL PURCHASE CARD

CIRCUIT DIAGRAM & PARTS REFERENCE / JST EDITION

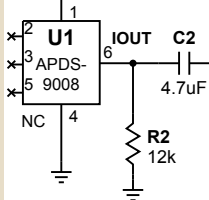
POWER



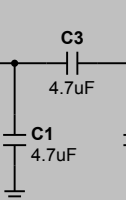
LED



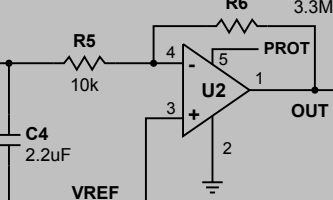
SENSE



FILTER



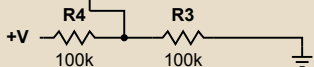
AMPLIFY



READING THE DRAWING

Solid lines are electrical connections.
A solid dot joins wires.
NC is unused.
Matching supply labels connect.

BIAS



BENCH REFERENCE

Keep this card beside the circuit board.

PARTS SCHEDULE

MARKING	VALUE / TYPE	PURPOSE
R1	LED current limiter	Limits LED current
R2	12k ohm	Current-to-voltage load
R3, R4	100k ohm each	Reference divider
R5	10k ohm	Amplifier input
R6	3.3M ohm	Amplifier feedback
C1, C3	4.7uF each	Filter network
C2	4.7uF	Input coupling
C4	2.2uF	Filter to reference
C5	2.2uF	Supply bypass
D1	Green LED	Illumination branch
D2	Schottky diode	Protected chip supply
U1	APDS-9008	Sensor chip
U2	MCP6001	Operational amplifier
J1	3-pin JST SH	Power and output

HOW TO FOLLOW THE CIRCUIT

U1 and R2 feed the capacitors. Follow the solid line to U2.

R6 returns OUT to the - input. VREF sets the + input midpoint.

The color tints match the functional areas on the PCB.

J1 pins: 1 GND (-) 2 +V 3 OUT